

OWC4DT: A 25 Gbps Plug-and-Play Self-Alignment Optical Wireless Communication System for Distributed Training in Data Center

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Abstract—Recently, the development of foundational models with massive parameters has increasingly relied on distributed training frameworks, which require synchronization of parameter gradients across multiple GPUs, raising the demand for high-speed interconnects within GPU clusters. Although high-bandwidth interconnect technologies like NVLink and InfiniBand support these distributed frameworks, they rely on wired media, leading to high cabling complexity and maintenance costs for large-scale data centers. To establish flexible and scalable data centers, Optical Wireless Communication (OWC) technology has emerged as a promising alternative to replace wired media. Most existing studies adopt the Fiber-Wireless-Fiber OWC (FWF-OWC) architecture, using custom-designed optical components for transceiver alignment and advanced techniques like wavelength division multiplexing (WDM) and high-order modulation to enhance communication rates. However, these systems are costly, require custom optical equipment, and have limited compatibility with optical modules. Additionally, current studies mainly conduct offline testing to measure physical-layer peak rates rather than real-time, end-to-end IP-level transmission rates, limiting practical applicability. To address these issues, we propose a minimalist FWF-OWC self-alignment system. This system performs alignment using a camera and a dual-axis stepper motor rotation platform, while being compatible with off-the-shelf SFP28 optical modules. Experimental results demonstrate that our system provides stable, plug-and-play IP-level connectivity over a 2-meter distance within a 10-degree field of view, achieving TCP and UDP transmission rates of 24.2 Gbps and 25.0 Gbps, respectively. Additionally, our system also maintains an alignment tolerance of ± 0.2 degrees at a distance of 2 meters, highlighting the potential for further cost reduction in the alignment mechanism.

I. INTRODUCTION

In recent years, with the rapid advancements of foundational models like Generative Pre-trained Transformer (GPT), the parameter scale of these models has expanded exponentially, far surpassing the memory capacity of a single GPU and making training tasks infeasible on a single device. To address this issue, academia and industry have developed distributed training frameworks, such as the Distributed Data Parallel (DDP) framework of PyTorch, which enable pre-training across large-scale GPU clusters. During training, model parameters and

datasets are partitioned across GPUs for parallel computation. The gradients of parameters, computed on each GPU, must then be broadcast and synchronized in real-time to ensure consistency and expedite convergence. As a result, distributed training frameworks demand high-speed interconnect links between GPUs, relying on technologies like NVLink and InfiniBand. However, these high-speed links are generally established using wired media, such as Printed Circuit Board (PCB) buses, Ethernet cables, coaxial cables, or fiber optics, which introduces significant complexity in cabling across large GPU server clusters and drives up both infrastructure costs and ongoing maintenance challenges for data centers.

Manufacturers have optimized GPU communication on the same motherboard through high-speed onboard circuits. However, when GPU units are housed in separate racks, significant loss in electrical signals over long distances makes optical fibers essential for reliable communication. With recent advancements in transmission power and reception sensitivity in optical modules, it is now possible to achieve stable fiber-optic communication over distances of 40 to 80 kilometers without amplifiers. Given these improvements, incorporating auto-alignment technology with optical modules at the transmitter and receiver can achieve precise axial alignment of the signal light, supporting a potential transition from wired optical networks to wireless configurations in data centers [1], [2]. This shift would enhance system scalability without compromising communication speeds. For instance, during data center expansion, Optical Wireless Communication (OWC) technology could simplify network integration by enabling direct alignment of transceivers, thereby reducing the engineering complexity and labor required for installing long-distance optical fibers. However, optical modules are primarily designed for fiber-optic communication, characterized by a narrow divergence angle and a reliance on collimator for precise coupling. Applying optical modules to free-space propagation requires further refinement in optical alignment to ensure that light signals are accurately coupled from free space into the optical fiber core on the transceiver [3].

To achieve high-precision alignment, a dual-axis rotation mechanism and corresponding dedicated automatic control

algorithm need to be designed. The authors of [4] introduced a novel auto-aligned optical wireless receiver for indoor mobile users, utilizing a dual-axis actuator, motion sensor, and gradient descent control algorithm to dynamically adjust and optimize received power, maintaining beam alignment as users move. Unlike the spatial multiplexing method used in [4], where Pencil Radiating Antennas (PRAs) transmit multiple optical pencil beams in different directions based on the beam's wavelength, [5] employed Wavelength Division Multiplexing (WDM) to further increase communication speed. Specifically, the authors of [5] combined the Surface Grating Coupler (SGC) and the Arrayed Waveguide Grating Router (AWGR) to achieve 4-WDM multiplexing over a distance of 10 centimeters, with each channel using 25 Gbps NRZ-OOK modulation, resulting in a total rate of 100 Gbps. Aside from using WDM technology, [6] employed Dual Polarization Quadrature Phase Shift Keying (DP-QPSK) modulation to increase the single-wavelength rate to 40 Gbps on inter-satellite links. In response to the demand for integration, [7] introduced an integrated device combining a 100 GHz uni-traveling carrier photodetector with a 100 GHz pHEMT amplifier, specifically designed for Fiber-Wireless-Fiber (FWF) communication systems exceeding 20 Gbps. This device incorporates a 2D Photodetector Array (2D-PDA) that employs 4-WDM in free-space communication, achieving 25 Gbaud per detector pixel across four pixels, resulting in a total data rate of 100 Gbps. In the context of data center applications, [8] proposed a novel OWC Data Center Network (OWC-DCN) architecture based on passive diffractive optics and fast tunable transmitters. Experimental validation demonstrated a 20 Gbps NRZ-OOK transmission rate in an 8x8 rack cluster, highlighting its potential to support high-speed communication within data centers with high scalability and low crosstalk. To enable the use of off-the-shelf optical communication products in the FWF system, such as SFP+ packaged optical modules, the authors of [9] connected the SFP+ optical module to a collimator via a single-mode fiber. The collimator then shaped the light from the fiber into a narrow parallel beam, which was directed through a combination of Fast Steering Mirrors (FSMs) and cameras, allowing for precise beam steering and alignment. With this setup, they achieved transmission rates of 20.6 Gbps for the downlink and 13.425 Gbps for the uplink.

The aforementioned studies have shown promising outcomes in terms of transmission rate and alignment precision. However, their broader industry adoption is hindered by two key factors. Firstly, the single-wavelength NRZ-OOK rates in these studies are capped at around 20 Gbps. To escalate transmission rates further, researchers have turned to techniques such as WDM, QPSK, and higher-order modulation. These approaches, while effective, require costly custom components, including modulators, signal amplifiers, optical antennas, and precise optical alignment structures at the transceiver, which prevents them from leveraging the benefits of mature, cost-effective, off-the-shelf optical modules. Secondly, rate testing in these studies is conducted offline rather than in a plug-and-play, end-to-end, IP-level setup. In these papers, a signal

generator produces random bits at the transmitter, which are modulated onto a laser diode. At the receiver, a detector performs photoelectric conversion, and the signal is recorded on an oscilloscope. A Bit Error Rate Tester (BERT) then compares the output of the oscilloscope with the original data from the signal generator to measure the Bit Error Rate (BER) and transmission rate. Although useful for theoretical testing, this setup lacks practicality as it is not compatible with network adapter and does not account for Forward-Error-Correction (FEC) overhead in real-world applications.

In response to these issues, we have designed a 25 Gbps low-cost plug-and-play OWC system. Specifically, our contributions are as follows:

- To reduce costs, we constructed a **minimalist FWF-OWC system** that comprises solely a dual-axis stepper motor rotation platform, an SFP28 optical module along with its collimator, a camera, a marker ring and an automatic self-alignment program. Notably, **we avoided using any custom optical components**, such as the PRAs in [4] and the SGC and AWGR in [5].
- We excluded WDM, QPSK, and higher-order modulation techniques, as these methods are uncommon in cost-effective optical modules, which typically employ simple NRZ-OOK modulation. Our NRZ-OOK encoding approach simplifies implementation without requiring complex modulation hardware. **For a fair comparison, we adjusted the peak rates in similar studies by their multiplexing gains**, given that their single-channel rates with NRZ-OOK were around 20 Gbps, and their rate improvements were achieved solely via multiplexing. Our system surpassed the baseline performance in comparable studies [9], achieving a peak physical layer transmission rate of **25 Gbps over 2 meters with single-wavelength NRZ-OOK**, demonstrating **excellent compatibility with off-the-shelf optical modules**.
- To augment the practical applicability, our system was crafted as a **plug-and-play IP-level transmission channel for servers** and was evaluated with two widely utilized transport layer protocols, **TCP** and **UDP**, achieving respective rates of **24.2 Gbps** and **25.0 Gbps**.

II. SYSTEM ARCHITECTURE

As shown in Fig. 1, we consider a data center containing N GPU server racks, with each rack equipped with an OWC self-alignment mechanism at the top of the rack to establish FWF links for high-speed communication between racks. Specifically, once the self-alignment mechanism completes optical alignment, the servers within the rack can connect to SFP28 optical modules via PCIe network adapters with optical ports. By simply configuring the IP addresses of these optical modules, IP networking between servers is achieved, making our system a plug-and-play IP-level transmission channel from the server's perspective. For clarity, we will focus on a single pair of transmitter and receiver self-alignment mechanisms in this paper, with the detailed design shown in the upper part of Fig. 1. Our system can be divided into three main subsystems:

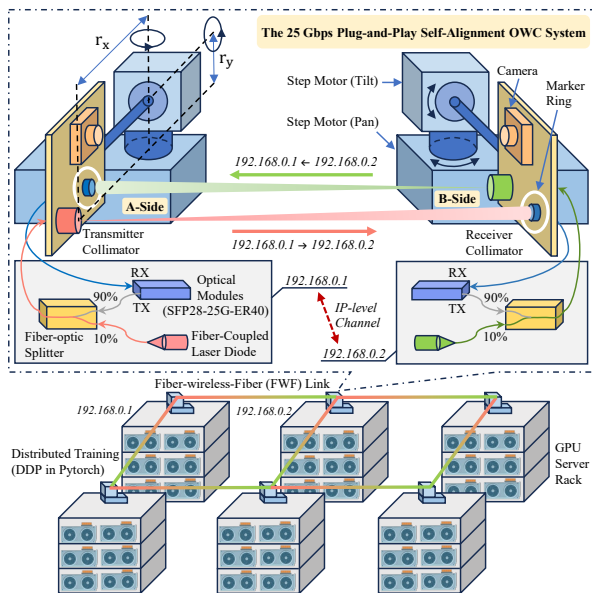


Fig. 1. Overall system architecture.

the optical communication system, the dual-axis mechanical rotation platform, and the self-alignment control system. The following three subsections will explain the implementation and principles of each subsystem.

A. Optical Communication System

Given that our system is based on an FWF-OWC architecture, we use SFP28 packaged optical modules as transceivers. Existing optical modules come in two types: single-fiber and dual-fiber. Single-fiber modules use WDM within a single fiber, typically utilizing wavelengths of 1310 nm for transmission and 1550 nm for reception, or vice versa. Dual-fiber modules, on the other hand, have separate TX and RX ports, both operating at the 1310 nm wavelength. Single-fiber modules using WDM face challenges in increasing transmission rates due to signal dispersion from the two different wavelengths, requiring dispersion compensation components, such as collimators with dispersion compensation, for coupling free-space light to the fiber. Additionally, single-fiber modules must be used in pairs, with complementary wavelength configurations at each end. For instance, if one end uses a TX1310/RX1550 nm module, the other end must use a TX1550/RX1310 nm module. However, dual-fiber modules avoid this issue by operating on a single wavelength band. Therefore, our system selects SFP28-25G-ER40 dual-fiber optical modules, which operate at 1310 nm and support a maximum fiber transmission distance of 40 km, with ER indicating Extended Range.

Since the 1310 nm signal light is in the near-infrared range, it cannot be captured by low-cost standard visible-light cameras. To enable the camera to track the beam position in real-time as feedback for the self-alignment system, we use a fiber-optic coupler to combine 90% of the 1310 nm signal light with 10% of a beacon light. The beacon light is produced by either a 525 nm green fiber-coupled laser diode or a 660 nm red fiber-coupled laser diode. The significant difference in wavelengths between the signal and beacon lights allows

the internal filter of the optical module to effectively block the beacon light, ensuring it functions solely as a marker for the invisible signal light's position and does not interfere with communication. The collimator converts the combined light into a narrow parallel beam for transmission. At the receiving end, the same type of collimator is used as an optical antenna to couple the free-space light into the fiber.

As shown in Fig. 1, it is important to note that the beacon lights on the A-side and B-side are different. From the camera's perspective, it can view not only the light spot from its own beam but also the interfering light spot emitted from the opposing side's beam. Therefore, we equip the cameras on the A-side and B-side with filters centered at 660 nm and 525 nm, matching their beacon lights. This configuration ensures that each camera observes only the light spot of its own beam while filtering out interference from the opposite side.

B. Dual-axis Mechanical Rotation Platform

In our system, beam alignment between the transmitter and receiver is achieved through a dual-axis mechanical rotation platform, allowing the beam emitted by the transmitter collimator to be received by the receiver collimator. The dual-axis rotation platform consists of two independent single-axis rotation platform, each driven by a stepper motor with a step angle of 1.8° . A worm gear transmission system with a 1/90 reduction ratio is used for speed reduction.

Additionally, the stepper motor driver is designed for a two-phase hybrid stepper motor, allowing the subdivision precision to be set by adjusting switches, with a maximum subdivision of 32. To interface the control system on the server, which is equipped with self-alignment control system, with the stepper motor driver, we use a control board based on the STM32F407/417, which converts the control commands from the server into pulse commands for the driver.

As for the rotation platform setup, we choose a 32-step subdivision, meaning each pulse signal causes the motor to rotate 0.05625° , meeting our requirements for precise control of the rotation. Furthermore, we set a $100 \mu s$ pulse duration for the motor drive, achieving a rotation speed of 6.25° per second to enhance alignment speed and maintain accuracy.

C. Self-alignment Control System

The self-alignment control system is a software program deployed on the server. It detects the current position of the light spot and the center of the marker ring, which corresponds to the center of the receiver's collimator, within pixel coordinates by analyzing images captured by the camera. Through a coordinate transformation, which will be detailed in Section III, the program calculates the angular offset between the light spot and the marker ring center in both tilt and pan dimensions based on their pixel offset. The offset angle acts as the error signal for the Proportional-Integral-Derivative (PID) algorithm, generating real-time rotation control signals sent to the driver to adjust the two-axis rotation platform's azimuth, aligning the light spot with the marker ring center and completing transmitter-receiver alignment.

III. IMAGE RECOGNITION BASED SELF-ALIGNMENT ALGORITHM

To achieve high-precision alignment between the transmitter and receiver beams, this section will establish an image recognition based self-alignment algorithm. First, we discuss the method for identifying the light spot and marker ring within camera images. Subsequently, we design a coordinate transformation algorithm to map the detected positions from pixel coordinate system to alignment error angles in the world coordinate system. Finally, to minimize the error angle and achieve precise alignment, we develop a stepper motor control algorithm based on the PID algorithm.

A. Image Recognition Algorithm

For the recognition of the marker ring, we utilize the Hough Circle Detection (HCD) algorithm rather than the more common deep learning-based image recognition algorithms. This choice is made because many deep learning algorithms achieve positioning by selecting from massive predefined anchor boxes, as in the case of YOLO, where the accuracy heavily relies on the density of anchor boxes. These algorithms cannot achieve pixel-level accuracy and tend to exhibit target box jitter due to changes in ambient lighting conditions. Conversely, the HCD algorithm is a geometry-based detection method, minimally impacted by image brightness or background complexity, making it ideal for high-precision circle detection. The main steps of the HCD algorithm are as follows:

- **Edge Detection:** Edge points in the image are extracted using Canny edge detection, capturing the potential circular contours.
- **Parameter Space Voting:** For each edge point, assuming it lies on a circle boundary, votes are cast in parameter space for various radius values. Specifically, potential circle centers are calculated for a range of assumed radii based on the edge point locations, with votes accumulated in parameter space for these center positions.
- **Peak Detection:** By scanning through the parameter space, points with peak votes are identified as the most probable circle centers and radii. These high-vote points represent the likely locations of circles in the image.
- **Filtering and Non-Maximum Suppression:** To further improve accuracy, non-maximum suppression is used to remove overlapping or incorrectly identified circles that do not meet the set radius or center distance criteria.

Additionally, the signal light and beacon light are mixed at a 9:1 power ratio and emitted from the collimator, then reflected by the marker ring of the receiver and captured by the camera, appearing as bright spots in the RGB channels of the image. By selecting the image from the corresponding wavelength channel and applying binarization and mean filtering, the pixel with the highest brightness value can be identified as the center coordinate of the light spot.

B. Azimuth Angle Estimation via Coordinate Transformation

Through image recognition, we can obtain the pixel positions of the centers of the light spot and the marker ring.

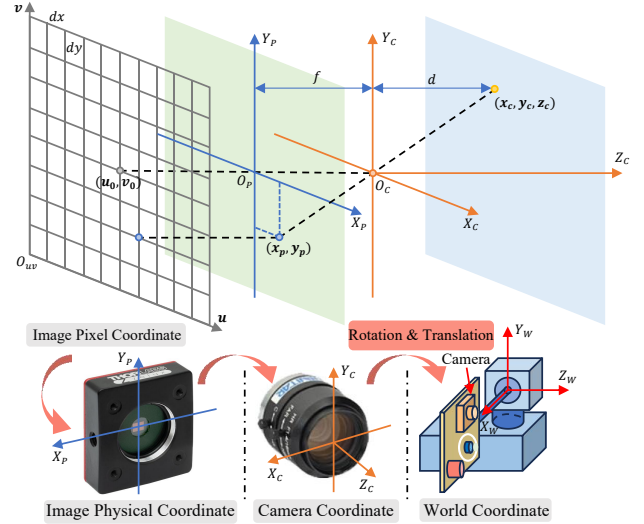


Fig. 2. The process of coordinate transformations.

These pixel positions must undergo a series of transformations to determine their coordinates in 3D space, which are then used to calculate the azimuth angle. As shown in Fig. 2, the coordinate transformations involve four coordinate systems: the image pixel coordinate system, the image physical coordinate system, the camera coordinate system, and the world coordinate system.

In the image pixel coordinate system, (u, v) are a series of discrete integer values representing pixel positions. By using the physical size of each pixel (d_x, d_y) , the image physical coordinates (x_p, y_p) can be calculated. However, the pixel coordinate system is established from one corner of the image, rather than from the center. Therefore, a correction is needed based on the pixel center coordinates (u_0, v_0) . The specific transformation is as follows:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{d_x} & 0 & u_0 \\ 0 & \frac{1}{d_y} & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix}. \quad (1)$$

The camera coordinate system is established with the optical center of the camera lens as the origin, with the x and y axes parallel to the image physical coordinate system and the z axis perpendicular to the image plane. The camera operates based on the pinhole imaging principle, so there is a similarity relationship between the object and its image. By considering the camera focal length f and the distance d from the optical center to the object, the following transformation formula can be derived as:

$$z_c \begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix}. \quad (2)$$

Finally, we choose the world coordinate system with the axes of the dual-axis rotation platform as the xy -axes, requiring the camera coordinate system to undergo rotation and translation to align with the world coordinate system. The specific steps are as follows:

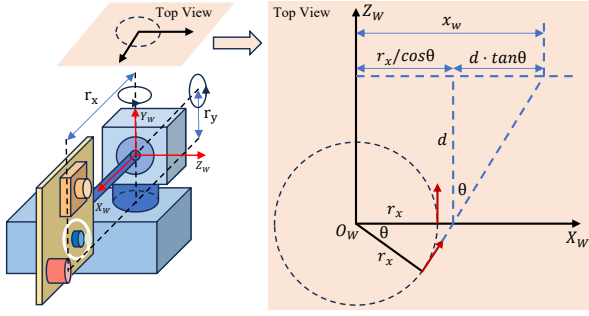


Fig. 3. Correction for azimuth angle due to transmitter collimator offset.

$$\begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} R_{3 \times 3} & t_{3 \times 1} \\ 0_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}, \quad (3)$$

where R represents an orthogonal unit rotation matrix, and t represents a three-dimensional translation vector. Note that we use homogeneous coordinates, which add an extra dimension to the coordinate vector. This allows all rotation and translation transformations to be represented as matrix multiplications, thereby unifying and simplifying the calculations. In summary, the coordinate transformation is expressed as:

$$z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{d_x} & 0 & u_0 \\ 0 & \frac{1}{d_y} & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} R_{3 \times 3} & t_{3 \times 1} \\ 0_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}. \quad (4)$$

After coordinate transformation, we obtain the 3D positions of the light spot and marker ring centers, enabling calculation of their azimuth angle θ . Due to the distance (r_x, r_y) between the collimator and motor rotation axis, a correction is required. The same correction applies to the marker ring center. For simplicity, Fig. 3 illustrates the azimuth angle calculation in the horizontal direction, with the formula as follows:

$$x_w = \frac{r_x}{\cos \theta} + d \tan \theta. \quad (5)$$

C. Automatic Control Algorithm of Step Motor

Using Equation (5), we can calculate the azimuth angle corresponding to the 3D coordinates in the world coordinate system. We take the angular difference between the light spot center and the marker ring center as the error signal, which is then processed by a PID controller to minimize this error. The PID controller is a feedback control system that reduces the error between the target value and the actual value through three components: proportional control adjusts the output based on the current error, integral control accumulates past errors to eliminate bias, and derivative control predicts the error trend to reduce overshoot, achieving precise and stable control. Specifically, the rotation angle α of the stepper motor can be calculated using the following formula:

$$\begin{aligned} \theta_{\text{error}} &= \theta_{\text{light spot}} - \theta_{\text{marker ring}}, \\ \alpha &= K_p \theta_{\text{error}} + K_i \int \theta_{\text{error}} dt + K_d \frac{d\theta_{\text{error}}}{dt}. \end{aligned} \quad (6)$$

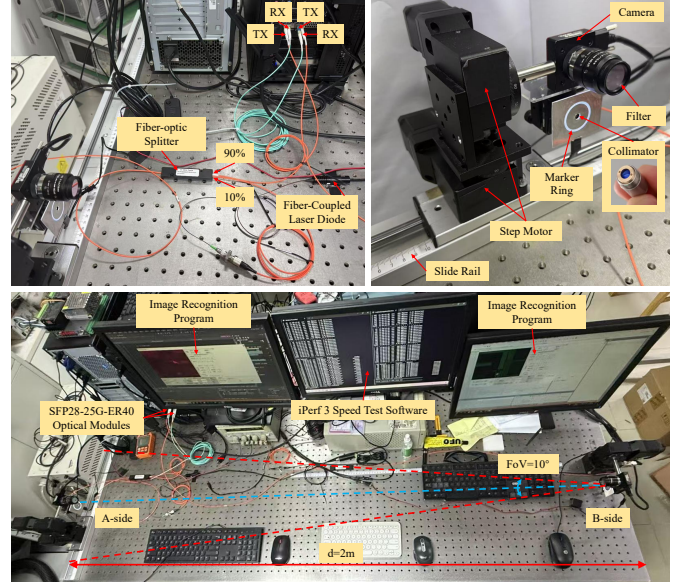


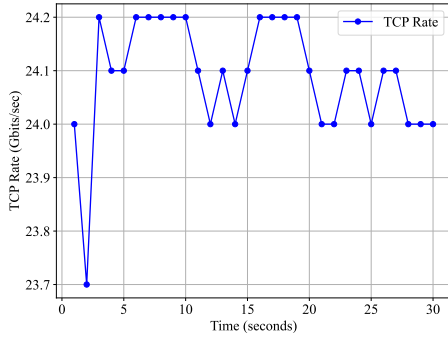
Fig. 4. The system setup of experimental validations.

IV. EXPERIMENTAL VALIDATIONS

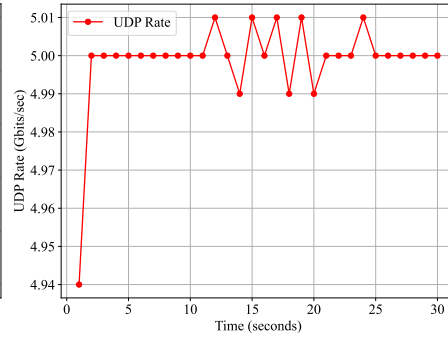
Constrained by limited equipment, we build a self-alignment OWC system on a 2-meter-long optical experiment platform, with the experimental setup shown in Fig. 4. We use dual-port SFP28-25G-ER40 optical modules, capable of achieving a physical layer transmission rate of 25.78 Gbps over a maximum fiber distance of 40 km. In our setup, the A-side transceiver is mounted on a sliding rail, while the B-side transceiver is fixed on the optical platform, with a straight-line distance of 2 meters between them. The sliding rail allows a maximum movement of 0.4 meters, corresponding to a Field of View (FoV) angle of 10 degrees.

The self-alignment program detailed in Section III is implemented in C# with a graphical user interface, as shown on the right side of Fig. 6. Unlike similar work, our system emphasizes constructing a plug-and-play, IP-level communication link. Therefore, we select iPerf3, a tool specifically designed to actively measure the maximum achievable bandwidth on IP networks, for our testing. To reach the peak communication rate of 25 Gbps, strong I/O performance is essential. Thus, we use a PC with an Intel 14700K CPU and 32GB of memory, equipped with an Intel Ethernet Network Adapter E810-XXVDA2. The adapter has two SFP28 ports, each connected to one of the optical modules. Both modules are set to different IP addresses within the same subnet, simulating communication between two remote servers.

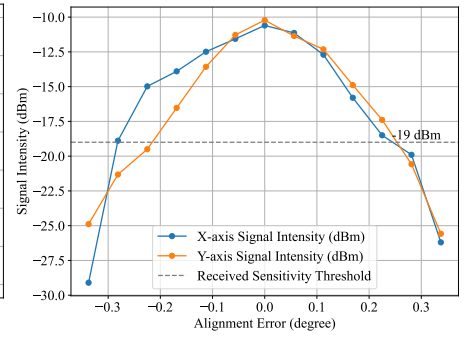
Specifically, we use the iPerf3 command in TCP mode with the "-P 10" option to establish 10 simultaneous connections for testing. The TCP rate remains consistently above 24 Gbps, with peak rates reaching 24.2 Gbps, as shown in Fig. 5(a). Through extensive testing, we find that the UDP mode of iPerf3 experiences significant packet loss at high throughput rates, even with the socket buffer size set to the maximum of 512 MB using the "-w 512M" option. To address this issue, we develop a Python script that initiates iPerf3 UDP tests across five separate pairs of command line windows, with each pair



(a) TCP rate test



(b) UDP rate test



(c) Signal intensity variation with alignment error

Fig. 5. System online experimental results.

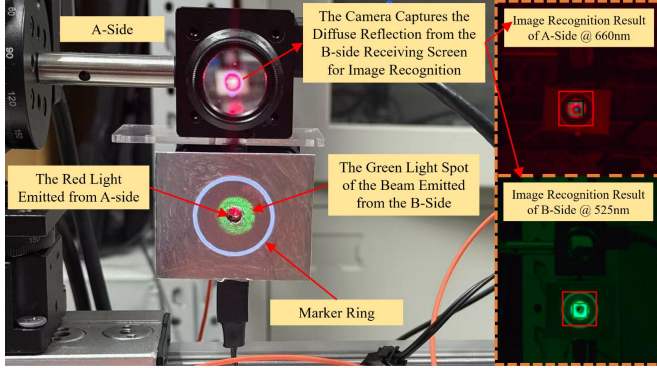


Fig. 6. Photograph of receiving screen after alignment and recognition results.

configured to a 5 Gbps UDP rate bound to a unique IP port. Fig. 5(b) shows the UDP test results for one of the connections, and the results for the other connections are nearly identical. Each individual UDP rate stabilizes at 5 Gbps, bringing the total UDP rate to 25 Gbps. The jitter ranges from 0.616 to 0.655 ms, with a packet loss rate of zero.

Additionally, Fig. 6 shows the alignment of the light spot and marker ring on the A-side receiving screen after the alignment process. It can be seen that the light spot is precisely coaxial with the collimator. Furthermore, we test the variation in signal light intensity coupled into the optical fiber through the collimator under non-coaxial conditions, as shown in Fig. 5(c). Specifically, after completing alignment, we introduce intentional alignment error via program control, scanning in the x and y directions at the minimum step angle of the stepper motor, which is 0.05625 degrees. From the Fig. 5(c), it is evident that due to certain manufacturing tolerances in the collimator, the beam profile is not a perfect Gaussian, exhibiting some asymmetry. The receiving sensitivity of our optical module is -19 dBm, which means our system can still function effectively with an angular misalignment of ± 0.2 degrees at a distance of 2 meters. In the case of a shorter communication distance, the tolerance for angular misalignment would be greater. This finding is significant because the minimum step angle of mechanical structures is closely linked to cost, and increased coaxial alignment tolerance could greatly reduce alignment mechanism costs. Therefore, our work has positive implications for the cost-effective implementation of FWF-OWC systems.

V. CONCLUSION

This paper presents a minimalist FWF-OWC system that aligns the transceiver axes using a camera to recognize light spot and marker ring positions, controlling a dual-axis rotation platform. Unlike previous offline-validation approaches, our system offers a plug-and-play IP-level channel, establishing a reliable 2-meter connection with a 10-degree FoV, achieving TCP speeds of 24.2 Gbps and UDP speeds of 25.0 Gbps. It also features a ± 0.2 -degree alignment tolerance at 2 meters, enabling cost reductions in mechanical design. The system could potentially be extended to the WDM domain using single-fiber optical modules and achromatic collimating lenses.

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